ALI ASHAR SHAFIQUE AI Engineer | Data Scientist | Machine Learning

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SUMMARY

AI/ML engineer who builds and ships LLM-powered applications end to end — from data ingestion and retrieval pipelines to FastAPI/React interfaces and containerized deployment. Built production-style RAG systems (FastAPI, ChromaDB, PostgreSQL, local + hosted LLMs) and an event-driven full-stack AI platform with function-calling. Strong in Python, SQL, and PyTorch/TensorFlow; comfortable owning a feature from prototype to deploy. Recent CS graduate seeking an AI Engineer role.

PROFESSIONAL EXPERIENCE

Machine Learning & Generative AI, Big Brains 08/2026 – 09/2026 | Lahore, Pakistan
- Researched and experimented with generative AI and LLM-based approaches, evaluating model outputs to inform implementation decisions on applied use cases.
- Built and optimized data preprocessing pipelines to clean and structure raw datasets for downstream machine learning workflows

Data Science Intern, Knowatech 08/2025 – 09/2025 | Lahore, Pakistan
- Engineered a crime reporting and analytics system processing **2,000+ FIR records**, improving complaint-tracking efficiency by **35%**.
- Cleaned and transformed **2,000+ rows of crime data** using Python (Pandas, NumPy) and Excel to surface crime trends and patterns for reporting.
- Developed Flask-based backend APIs powering data visualization dashboards and automated reporting workflows used by the analytics team.

Data Science Intern, Dev Hub 03/2025 – 05/2025 | Lahore, Pakistan
- Conducted exploratory data analysis, feature engineering, and predictive modeling using Python and SQL to support business forecasting
- Designed automated dashboards tracking KPIs, cutting manual reporting time and giving stakeholders real-time visibility into strategic metrics.

Information Systems Intern, Fauji Fertilizer Company Ltd. 07/2024 – 08/2024 | Lahore, Pakistan
- Supported enterprise systems spanning IT, HR, Finance, and Sales departments across the organization.
- Managed and audited **1,000+ internal records**, streamlining data workflows and improving record accuracy.

PROJECTS

Nigehban — Real-Time Crime Monitoring Platform
- Designed and built a full-stack crime intelligence dashboard tracking **4,200+ FIR records, 364 officers, and 36 police stations** across a 6-level geographic hierarchy, using an async FastAPI backend with SQLAlchemy 2.0 and PostgreSQL
- Built a real-time WebSocket incident feed with a background scheduler generating statistically weighted events every 8 seconds, based on historical crime-type distributions
- Developed a React 19 + TypeScript frontend with Leaflet mapping, TanStack Query, and Recharts-based dashboards across 6+ live KPIs
- Designed a 5-table relational PostgreSQL schema with Alembic migrations and Dockerized the full stack for one-command setup

Investigation Intelligence Platform
- Architected a full-stack RAG app — React chat UI, FastAPI backend, PostgreSQL + ChromaDB, and a locally-hosted LLM for generation
- Built a FastAPI REST API (Pydantic validation) connecting the chat UI to a retrieval/generation pipeline, returning cited, source-grounded answers
- Built a data ingestion pipeline (API integration, OCR, NLP entity extraction) to structure unstructured case records for retrieval
- Implemented multi-thread conversation history, expandable source citations, and backend-readiness polling
- Dockerized the full stack for one-command, reproducible local deployment

Bizionary – AI-Enabled ERP System for SMEs ,
Full-stack, decoupled business intelligence and resource-planning suite with an event-driven architecture
- Built a React (Vite) dashboard with Tailwind CSS and Recharts to manage sales, invoices, and real-time financial tracking for SME operations.
- Designed stateless Django REST APIs with JWT authentication and a Pandas-driven Excel parser to ingest and process monthly financial reports.
- Implemented Django Signals to trigger real-time, atomic updates to inventory ledgers and double-entry charts of accounts, preserving transactional integrity.
- Embedded a Groq-powered (Llama 3.3) chatbot using function-calling to query live database state and forecast demand for business users.

Automated ETL Pipeline with Apache Airflow and Real-Time Analytics Dashboard
End-to-end data engineering project
- Built a fully automated ETL pipeline processing sales records with Python, Pandas, and SQLite, orchestrated on a scheduled basis via Apache Airflow.
- Developed an interactive Streamlit dashboard visualizing sales trends, top products, and annual metrics for business stakeholders.
- Performed data cleaning, feature engineering, and aggregation to generate actionable business insights from raw sales data.

SKILLS

Languages & Libraries: Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch), SQL
Machine Learning / AI: RAG systems, NLP, LLM integration (Groq, Llama 3.3), Feature Engineering, OCR (Tesseract)
Backend & Web: FastAPI, Flask, Django, REST APIs, React
Data Engineering: Apache Airflow, ETL pipelines, PostgreSQL, SQLite, Data Warehousing
Tools & Platforms: Docker, Docker Compose, Git, Power BI, Tableau, Jupyter

EDUCATION

University of Management and Technology, Bachelor's: Computer Science 08/2022 – 07/2026

Government College University, Intermediate: Computer Science 01/2019 – 08/2021

CERTIFICATIONS

IBM: Data Science Professional Certificate - Intro to DevOps and GEN AI
Deep Learning.AI: Linear Algebra for ML, Calculus for ML, AI for Everyone
NVIDIA Whizlabs: Large Language Models and AI Deployment
Deloitte: Data Analytics Job Simulation